

## 35.5 Case-control sampling

A rare disease affects one person in a thousand, so a prospective study hoping to see a hundred cases must follow a hundred thousand people, most of whom carry almost none of the weight  $\pi_i(1 - \pi_i)$ . The economy is to go looking for the cases. In a **retrospective**, or **case-control**, design the response decides who enters the sample and the covariates are observed afterwards. This inverts the conditioning, and one might expect nothing to be estimable. In fact, for the logit link and no other, almost everything is.

### 35.5.1 Only the intercept moves

#### Theorem 35.5.1 — Retrospective sampling under a logistic model

Let  $(\mathbf{X}, Y)$  follow  $\text{logit } \Pr(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$  in the population, and let  $S \in \{0, 1\}$  indicate inclusion in the sample, with

$$\Pr(S = 1 \mid Y = y, \mathbf{X} = \mathbf{x}) = \rho_y \in (0, 1], \quad y = 0, 1,$$

depending on  $y$  but not on  $\mathbf{x}$ . Then:

- (a)  $\text{logit } \Pr(Y = 1 \mid \mathbf{X} = \mathbf{x}, S = 1) = \alpha + \log(\rho_1/\rho_0) + \mathbf{x}^\top \boldsymbol{\beta}$ . The slopes are unchanged; only the intercept moves, and it moves by a constant that does not involve  $\mathbf{x}$ ;
- (b) let  $F$  be a continuous strictly increasing distribution function, suppose  $\Pr(Y = 1 \mid \mathbf{x}) = F(\alpha + \mathbf{x}^\top \boldsymbol{\beta})$ , and suppose that for every ratio  $r = \rho_1/\rho_0$  in some open interval containing 1 there is a constant  $c(r)$  with  $\Pr(Y = 1 \mid \mathbf{x}, S = 1) = F\{\alpha + c(r) + \mathbf{x}^\top \boldsymbol{\beta}\}$  for all  $\mathbf{x}$ . If the linear predictor  $\alpha + \mathbf{x}^\top \boldsymbol{\beta}$  ranges over an interval  $J$  with nonempty interior as  $\mathbf{x}$  varies, then  $F(u) = \{1 + e^{-(au+b)}\}^{-1}$  for every  $u \in J$  and some  $a > 0$ ,  $b$ : wherever the population identifies it, the link is logistic;
- (c) if  $\rho_1$  and  $\rho_0$  are known, or if the population prevalence  $\bar{\pi} = \Pr(Y = 1)$  is known and the sample contains  $n_1$  cases and  $n_0$  controls, then  $\alpha$  itself is estimable: in the second case  $\hat{\alpha} = \hat{\alpha}_S - \log\{(n_1/n_0)(1 - \bar{\pi})/\bar{\pi}\}$ , where  $\hat{\alpha}_S$  is the intercept fitted to the sample.

*Proof.*

- (a) By Bayes' theorem, writing  $\pi = \Pr(Y = 1 \mid \mathbf{x})$ ,

$$\Pr(Y = 1 \mid \mathbf{x}, S = 1) = \frac{\rho_1 \pi}{\rho_1 \pi + \rho_0 (1 - \pi)},$$

so that

$$\frac{\Pr(Y = 1 \mid \mathbf{x}, S = 1)}{\Pr(Y = 0 \mid \mathbf{x}, S = 1)} = \frac{\rho_1}{\rho_0} \cdot \frac{\pi}{1 - \pi}.$$

Taking logarithms and using  $\text{logit } \pi = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$  gives the claim. The step needs  $\rho_y$  to be free of  $\mathbf{x}$ , and nothing else.

- (b) Put  $G = \text{logit} \circ F$ , a continuous strictly increasing function, and write  $\eta = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$ . The computation in (a) used only Bayes' theorem, so it holds for any  $F$ :  $\text{logit } \Pr(Y = 1 \mid \mathbf{x}, S = 1) = \log r + G(\eta)$ . The hypothesis says this equals  $G\{\eta +$

$c(r)\}$ . Hence

$$G\{\eta + c(r)\} - G(\eta) = \log r \quad \text{for all } \eta \text{ in an interval } J.$$

Since  $G$  is strictly increasing,  $c$  is strictly increasing in  $r$  with  $c(1) = 0$ , and as  $r$  runs over an open interval containing 1,  $v = c(r)$  runs over an open interval  $V$  containing 0. The displayed identity says that  $\kappa(v) = G(\eta + v) - G(\eta)$  is the same for every  $\eta \in J$  with  $\eta + v \in J$ . Taking  $v, v'$  and  $\eta$  with  $v, v', v + v' \in V$  and  $\eta, \eta + v, \eta + v + v'$  all in  $J$ , which the nonempty interior of  $J$  permits, gives  $\kappa(v + v') = \kappa(v) + \kappa(v')$ ;  $\kappa$  is continuous, so  $\kappa(v) = av$  near 0, with  $a > 0$  by monotonicity. Fix  $\eta_0$  in the interior of  $J$ . Then  $G(\eta) = G(\eta_0) + a(\eta - \eta_0)$  for  $\eta \in J$  within  $V$  of  $\eta_0$ , and chaining such steps along the interval  $J$  extends the identity to all of  $J$ . Inverting,  $F(u) = \text{logit}^{-1}(au + b)$  on  $J$ : nothing pins  $F$  down outside  $J$ , and nothing can, since the sampling scheme says nothing about linear predictors the population never produces.

- (c) By (a) the fitted intercept estimates  $\alpha + \log(\rho_1/\rho_0)$ , so knowing the ratio recovers  $\alpha$ . If instead  $\bar{\pi}$  is known and the population has  $N$  members, then  $\rho_1 = n_1/(N\bar{\pi})$  and  $\rho_0 = n_0/\{N(1 - \bar{\pi})\}$  in expectation, so  $\rho_1/\rho_0 = (n_1/n_0)(1 - \bar{\pi})/\bar{\pi}$ .

■

Part (a) is why the odds ratio dominates epidemiology. It is not that odds ratios are more natural than risk ratios — they are less natural — but that they are the only effect measure a retrospective study can estimate without knowing how the sample was drawn. Part (b) says this belongs to the logit alone.

*Remark* (Why the prospective likelihood is legitimate). Part (a) says the *model* transfers, not that maximizing the ordinary prospective likelihood on retrospective data is a correct procedure: the actual likelihood of such a sample is  $\prod_i f(\mathbf{x}_{(i)} \mid y_i)$ , which involves the unknown population distribution of  $\mathbf{X}$ . Prentice and Pyke (1979) settled this. Treating that distribution nonparametrically and profiling it out of the retrospective likelihood yields exactly the prospective logistic likelihood, so the ordinary fit is the semiparametric maximum likelihood estimate of  $\boldsymbol{\beta}$  and the usual inverse information is its correct asymptotic covariance. Their argument profiles an infinite-dimensional nuisance parameter and is not reproduced here.

### Example. (A simulated retrospective study)

Take a population of 200000 with a continuous exposure  $x_1 \sim N(0, 1)$ , an independent binary covariate  $x_2$ , and  $\text{logit } \pi = -5 + 1.2x_1 + 0.8x_2$ . The prevalence is 0.0178, so there are 3565 cases, and fitting the whole population gives  $\hat{\alpha} = -4.959$ ,  $\hat{\beta}_1 = 1.202$ ,  $\hat{\beta}_2 = 0.748$ .

Now take every case and four random controls per case, and fit the same model to those records as if they had been collected prospectively. Averaged over 400 control samples the slopes are recovered, at  $\hat{\beta}_1 = 1.201$  and  $\hat{\beta}_2 = 0.755$ , while the intercept averages  $-2.339$  against the predicted  $\hat{\alpha} + \log(\rho_1/\rho_0) = -4.959 + 2.623 = -2.337$ .

With data generated from a *probit* model and fitted with a probit link, the population slope is 0.547 while the retrospective samples average 0.719, too large by a factor of 1.31 (Figure 35.5.1).

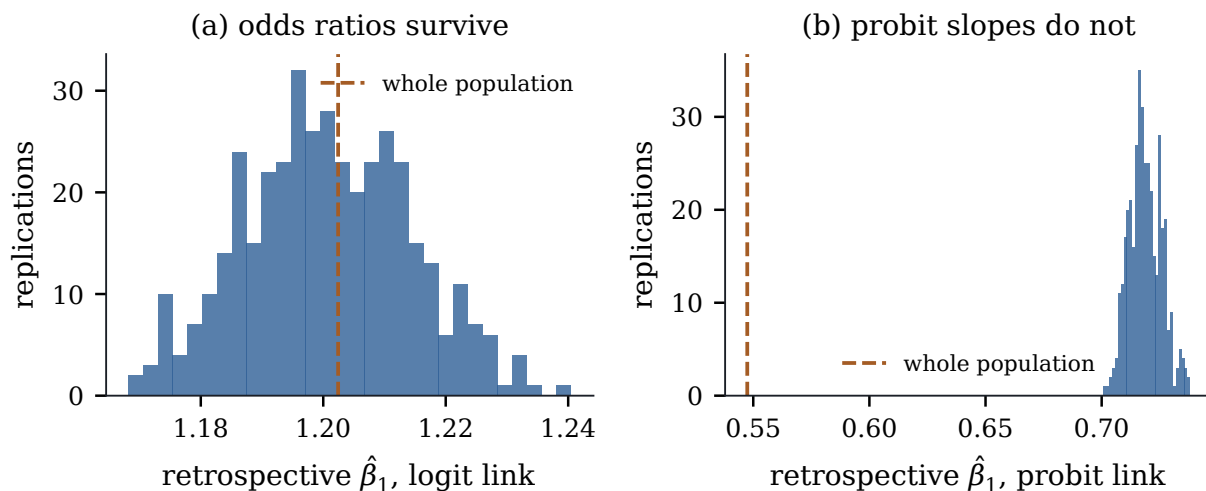


Figure 35.5.1: Retrospective estimates of a slope from 400 case-control samples, with the whole-population estimate marked. (a) Logistic data under the logit link: centred on the population value. (b) Probit data under the probit link: systematically too large.

```
import numpy as np
import statsmodels.api as sm

rng = np.random.default_rng(20)
N = 200_000
alpha, beta1, beta2 = -5.0, 1.2, 0.8
x1 = rng.normal(size=N)
x2 = (rng.random(N) < 0.3).astype(float)
Xpop = np.column_stack([np.ones(N), x1, x2])
ypop = (rng.random(N) < 1 / (1 + np.exp(-(Xpop @ [alpha, beta1, beta2])))).astype(float)

prospective = sm.GLM(ypop, Xpop, family=sm.families.Binomial()).fit()
cases = np.flatnonzero(ypop == 1)
controls = np.flatnonzero(ypop == 0)
n1 = len(cases)

print("prevalence", ypop.mean().round(4), " cases", n1)
print("whole population:", prospective.params.round(3))
```

[Run this code online](#)

```
def case_control(rng, ratio=4, link=None):
    """Take every case and `ratio` controls per case; fit the same model
    prospectively."""
    keep = np.r_[cases, rng.choice(controls, ratio * n1, replace=False)]
    family = sm.families.Binomial(link=link) if link else sm.families.Binomial()
    return sm.GLM(ypop[keep], Xpop[keep], family=family).fit().params

rng2 = np.random.default_rng(7)
draws = np.array([case_control(rng2) for _ in range(400)])
shift = np.log(1.0 / (4 * n1 / len(controls))) # log(rho_1 / rho_0)

print("mean case-control estimate:", draws.mean(0).round(3))
print("predicted intercept:", round(prospective.params[0] + shift, 3))
```

```
print("slopes in the population:", prospective.params[1:].round(3))
```

[Run this code online](#)

## 35.5.2 Matched designs

Controls are often matched to cases on variables that matter but are not of interest: age, sex, neighbourhood. A matched set is one case and  $M$  controls sharing those variables, and the model gives each set its own intercept:

$$\text{logit } \Pr(Y_{kj} = 1 \mid \mathbf{x}_{kj}) = \alpha_k + \mathbf{x}_{kj}^\top \boldsymbol{\beta}, \quad k = 1, \dots, K, \quad j = 0, 1, \dots, M.$$

Estimating the  $K$  intercepts along with  $\boldsymbol{\beta}$  is the incidental-parameter situation of Example (Many means, one variance), in which maximum likelihood is not consistent. Conditioning removes them.

### Proposition 35.5.2 — Conditional logistic regression

In the model above, suppose exactly one member of each set is a case, and condition on that fact. Then:

- (a) the conditional probability that member  $j^*$  of set  $k$  is the case, given the covariates of the set and that exactly one member is, is

$$\frac{\exp(\mathbf{x}_{kj^*}^\top \boldsymbol{\beta})}{\sum_{j=0}^M \exp(\mathbf{x}_{kj}^\top \boldsymbol{\beta})}, \quad (35.5.1)$$

free of  $\alpha_k$ ;

- (b) the product of these over  $k$  is the **conditional likelihood**; maximizing it gives a consistent, asymptotically normal estimate of  $\boldsymbol{\beta}$  as  $K \rightarrow \infty$  with  $M$  fixed, and the ordinary inverse-information standard errors apply to it;
- (c) for  $M = 1$  (matched pairs), let  $\mathbf{d}_k = \mathbf{x}_{k1} - \mathbf{x}_{k0}$  be the case-minus-control difference in a discordant pair. The conditional log-likelihood is  $-\sum_k \log\{1 + \exp(-\boldsymbol{\beta}^\top \mathbf{d}_k)\}$ , while the profile log-likelihood obtained by maximizing the *unconditional* likelihood over the  $\alpha_k$  is  $-2\sum_k \log\{1 + \exp(-\frac{1}{2}\boldsymbol{\beta}^\top \mathbf{d}_k)\}$ . Consequently the unconditional estimate is exactly twice the conditional one, and converges to  $2\boldsymbol{\beta}$ .

*Proof.*

- (a) With  $\pi_{kj} = \text{logit}^{-1}(\alpha_k + \mathbf{x}_{kj}^\top \boldsymbol{\beta})$ , the probability that  $j^*$  alone is a case is  $\pi_{kj^*} \prod_{j \neq j^*} (1 - \pi_{kj})$ . Dividing by the sum of the same expression over all choices, and cancelling  $\prod_j (1 - \pi_{kj})$  from numerator and denominator, leaves  $\{\pi_{kj^*} / (1 - \pi_{kj^*})\} / \sum_j \{\pi_{kj} / (1 - \pi_{kj})\}$ . Each odds is  $e^{\alpha_k} e^{\mathbf{x}_{kj}^\top \boldsymbol{\beta}}$ , and  $e^{\alpha_k}$  cancels, giving [Eq. 35.5.1](#).
- (b) The sets are independent and [Eq. 35.5.1](#) is a genuine probability model for a fixed parameter  $\boldsymbol{\beta}$  of fixed dimension, so the standard likelihood asymptotics apply as  $K \rightarrow \infty$ . (It is the likelihood of a multinomial logit model, which [Chapter 36](#) studies in its own right.)
- (c) Part (a) with  $M = 1$  gives  $\{1 + \exp(-\boldsymbol{\beta}^\top \mathbf{d}_k)\}^{-1}$ , whose logarithm summed over  $k$  is the stated conditional log-likelihood. For the unconditional calculation, write

$u = \alpha_k + \frac{1}{2}\boldsymbol{\beta}^\top(\mathbf{x}_{k1} + \mathbf{x}_{k0})$  and  $c = \frac{1}{2}\boldsymbol{\beta}^\top\mathbf{d}_k$ , so the two linear predictors are  $u + c$  and  $u - c$ . The pair's log-likelihood is  $(u + c) - \log(1 + e^{u+c}) - \log(1 + e^{u-c})$ ; differentiating in  $u$  gives  $1 - \text{logit}^{-1}(u + c) - \text{logit}^{-1}(u - c) = 0$ , that is,  $\text{logit}^{-1}(u + c) = \text{logit}^{-1}(c - u)$ , whose unique solution is  $u = 0$ . Substituting  $u = 0$ ,

$$c - \log(1 + e^c) - \log(1 + e^{-c}) = -2\log(1 + e^{-c}),$$

which is the stated profile contribution with  $c = \frac{1}{2}\boldsymbol{\beta}^\top\mathbf{d}_k$ . A concordant pair contributes a supremum of 0, approached as  $\alpha_k \rightarrow \pm\infty$  and free of  $\boldsymbol{\beta}$ . The profile log-likelihood is therefore twice the conditional log-likelihood evaluated at  $\boldsymbol{\beta}/2$ , so its maximizer is twice the conditional maximizer; since the latter is consistent for  $\boldsymbol{\beta}$ , the former converges to  $2\boldsymbol{\beta}$ . ■

### Example. (Doubling the effect)

Simulate 3000 matched pairs with one covariate, stratum intercepts drawn from a normal distribution and  $\beta = 1$ , keeping only the discordant pairs. The conditional estimate is 0.942 with standard error 0.037. Profiling the unconditional likelihood numerically over the three thousand intercepts gives 1.884, twice that to three decimals as [Proposition 35.5.2\(c\)](#) requires, and about twenty-four standard errors from the truth.

```
from scipy.optimize import minimize

def matched_pairs(K=3000, beta=1.0, seed=11):
    """K strata, each with one case and one control, and a stratum-specific
    intercept."""
    rng = np.random.default_rng(seed)
    a = rng.normal(-1.0, 2.0, K) # the nuisance intercepts
    xc, xk = np.empty(K), np.empty(K)
    for k in range(K):
        while True: # sample the stratum until
            it is discordant
            x = rng.normal(size=2)
            pr = 1 / (1 + np.exp(-(a[k] + beta * x)))
            yy = rng.random(2) < pr
            if yy[0] != yy[1]:
                xc[k], xk[k] = (x[0], x[1]) if yy[0] else (x[1], x[0])
                break
    return xc, xk # exposure of the case, of
                 the control

x_case, x_control = matched_pairs()
# the conditional likelihood for 1:1 matching depends on beta only through the
# difference
d = x_case - x_control
cond = minimize(lambda b: np.sum(np.log1p(np.exp(-b[0] * d))), [0.0]).x[0]
print("conditional estimate:", round(cond, 3))
```

[Run this code online](#)

Matching buys precision when the matching variables are strong confounders, and costs all information about them, since they are absorbed into the  $\alpha_k$ . Overmatching, matching on a variable on the causal path from exposure to outcome, removes part of the effect one is measuring: a bad control (Theorem 25.5.1).

### 35.5.3 Two warnings about odds ratios

#### Example. (The odds ratio does not collapse)

Generate a population of 200000 with two *independent* covariates,  $x_1$  standard normal and  $x_2$  Bernoulli with probability 0.3, and logit  $\pi = 1.0x_1 + 4.0x_2$ , giving a prevalence of 0.638. Fitting both covariates and an intercept recovers the coefficient of  $x_1$  as 1.006. Fitting an intercept and  $x_1$  alone, and thereby averaging over  $x_2$ , gives 0.745, a quarter smaller. There is no confounding here: the two covariates are independent by construction, and the marginal and conditional odds ratios simply differ.

The odds ratio is not *collapsible*: the best logistic approximation to an average of logistic curves is flatter than they are, by an amount growing with the variance of the omitted term ([Exercise \(C1\)](#)). Conditional and marginal odds ratios therefore answer different questions, and adding a covariate moves the other coefficients away from zero even when it is independent of them, so the reasoning “the coefficient changed, therefore the new variable is a confounder” is invalid here. Risk differences and risk ratios do collapse.

The second warning is causal. [Theorem 35.5.1](#) requires selection to depend on  $Y$  alone; if recruitment also depends on the exposure, say because the cases come from a hospital serving an exposed population, then  $S$  is a collider on a path between  $\mathbf{X}$  and  $Y$  and conditioning on  $S = 1$  opens it (Proposition 25.2.1). And the conditional odds ratio it estimates is the causal effect only if the covariates satisfy the backdoor criterion (Definition 25.4.1 and Theorem 25.4.1), which no sampling scheme can supply.

*Idea.* Retrospective sampling is the one design in which the choice of link is forced: [Theorem 35.5.1](#) makes the odds ratio estimable under a scheme that destroys every other effect measure. The price is that odds ratios do not collapse, do not approximate risk ratios unless the outcome is rare, and are causal only under the usual assumptions.

### 35.5.4 Exercises

#### 35.5.4.1 A. Check your understanding

##### Exercise A1.

A case-control study has 200 cases and 600 controls, and the disease affects 2% of the population. The fitted intercept is  $-0.8$ . Estimate the population intercept, and the probability of disease at the reference covariate values.

*Solution.* By [Theorem 35.5.1\(c\)](#),  $\log\{(200/600)(0.98/0.02)\} = \log(16.333) = 2.793$ , so  $\hat{\alpha} = -0.8 - 2.793 = -3.593$  and the probability is  $1/(1 + e^{3.593}) = 0.0268$ , close to the population prevalence as it should be.

## 35.5.4.2 B. Practice

**Exercise B1.**

Suppose  $\Pr(S = 1 \mid Y = y, \mathbf{X} = \mathbf{x}) = \rho_y g(\mathbf{x})$  for some  $g$  with values in  $(0, 1]$ . Show that [Theorem 35.5.1\(a\)](#) still holds. Then suppose instead that cases are sampled at rate  $\rho_1 e^{\mathbf{x}^\top \gamma}$  while controls are sampled at a rate free of  $\mathbf{x}$ ; find the bias in the estimated slopes.

**Exercise B2.**

For matched pairs with a single binary exposure, let  $n_{10}$  be the number of pairs with an exposed case and unexposed control and  $n_{01}$  the reverse. Show that the conditional estimate of the log odds ratio is  $\log(n_{10}/n_{01})$ , and that the conditional test of  $\beta = 0$  is a binomial test of  $1/2$  in  $n_{10} + n_{01}$  trials.

*Solution.* With a binary exposure  $d_k \in \{-1, 0, 1\}$ , and pairs with  $d_k = 0$  contribute a constant. The conditional log-likelihood is  $-n_{10} \log(1 + e^{-\beta}) - n_{01} \log(1 + e^{\beta})$ ; setting the derivative to zero gives  $e^{\beta} = n_{10}/n_{01}$ . Writing  $p = e^{\beta}/(1 + e^{\beta})$ , the conditional likelihood is that of  $n_{10}$  successes in  $n_{10} + n_{01}$  Bernoulli trials, and  $\beta = 0$  is  $p = 1/2$ . The resulting test is McNemar's, its exact version the binomial test.

**Exercise B3.**

With  $n_1$  cases fixed and  $n_0 = Mn_1$  controls, show that for a single binary exposure and a rare disease the asymptotic variance of the log odds ratio is proportional to  $1 + 1/M$ , and deduce that there is little gain beyond  $M = 4$  or  $5$ .

## 35.5.4.3 C. Going deeper

**Exercise C1.**

Let  $\text{logit} \Pr(Y = 1 \mid X, Z) = \beta X + \gamma Z$  with  $Z$  independent of  $X$ . Show that  $\Pr(Y = 1 \mid X)$  is not of the form  $\text{logit}^{-1}(\beta^* X)$ , and explain why the best logistic approximation has  $|\beta^*| < |\beta|$ . What happens as  $\gamma \rightarrow 0$ ?

*Solution.*  $\Pr(Y = 1 \mid X = x) = E_Z \text{logit}^{-1}(\beta x + \gamma Z)$  is a mixture of horizontally shifted logistic curves, and a nondegenerate such mixture is not itself logistic: its derivative at its median is strictly smaller than that of a single component with the same slope, by concavity of the response function above the median and convexity below. The marginal curve is flatter, so the best logistic fit has a smaller slope in modulus. As  $\gamma \rightarrow 0$ ,  $\beta^* \rightarrow \beta$ ; the attenuation grows with  $\gamma^2 \text{Var } Z$ , which is why the effect in [Example \(The odds ratio does not collapse\)](#) is large.